

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
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(1) Research Program/Project/Study Title:	Development of Instant Sopa de Fedio (Noodle Soup)
Proponent Name:	Dr. Ace D. Barredo
Academic Rank:	none
Co-Authors (with academic rank):	none
Campus/Satellite College/Address:	CapSU – Main Campus
Telephone/Cell Phone/Fax Number:	09994044417
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	Sopa de Fedio (Noodle Soup) Ala Arroz Caldo Sopa de Fedio (Noodle Soup) Ala Chicken Tinola
Component Leader:	none
Academic Rank:	none

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:	Roxas City						
Province:	Capiz						
Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity Agriculture/Fruit and Vegetables	(8) Discipline/Field of Specialization Food and Development						

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Instant noodles are a type of pre-cooked noodle usually sold in individual packets, cups, or bowls. Its main ingredients are typically flour, starch, water, salt, and/or a salt substitute known as kansui, a type of alkaline mineral water that contains sodium carbonate and usually potassium carbonate. Palm oil is also a common ingredient in instant noodles as instant noodles were originally produced by flash frying. However, nowadays, air-dried noodles are also available. Instant noodles come with flavoring packets that contain seasoning, salt, and monosodium glutamate (MSG). As much as people love consuming instant noodles for their convenience, affordability, and taste, many don't know that there are underlying health risks involved. Consuming instant noodles on a daily basis can have a number of very serious consequences for your health (parkwayeast.com.sg/healthplus/article/instant-noodles-health-impact).

However, since we're in a fast-paced kind of living many things are done in so easiest way even in the food preparation and the food we eat. We neglect the importance of eating healthy and nutritious foods, especially for young children. That's why many chronic diseases are linked to unhealthy foods and unhealthy lifestyles.

As a researcher, I would like to develop a healthy yet affordable instant noodle dish product with a twist on the common Filipino dish which is the "Arroz Caldo and Chicken Tinola" with the integration of vegetables that would be better than usual instant noodles. It will also answer the nutrient deficiencies among children and adults prevalent in the country. These concepts formed the bases for the conduct of the current study.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The proposed project focuses on the Development of Sopa de Fedio (Noodle Soup) Ala Arroz Caldo and Ala Chicken Tinola.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives: The general objective is to a healthy yet affordable noodle dish product with a twist on the common Filipino dish which is the "Arroz Caldo and Chicken Tinola" with the integration of vegetables. Specifically, it seeks to;

1. evaluate the sensory qualities of sopa de fedio ala arroz caldo and ala chicken tinola in terms of appearance, aroma, taste and texture.
2. determine the shelf-life of the sopa de fedio ala arroz caldo and ala chicken tinola in terms of time and temperature.
3. determine the product cost of sopa de fedio ala arroz caldo and ala chicken tinola.
4. find out the nutritional value of sopa de fedio ala arroz caldo and ala chicken tinola.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

The present study will be based on the following prior arts such as:

NAGAYAMA YOSHIKI +SANYO FOODS CO LTD + NZ733059 (A). Dried noodles and production method thereof. The purpose of the present invention is to produce instant swollen dried noodles in which "cracking of noodle strings", which has been a problem accompanying high-temperature hot-air drying, is prevented or suppressed. Disclosed are instant swollen dried noodles including: a main ingredient comprising wheat flour; and at least one cross-linked pregelatinized potato starch selected from the group consisting of etherified, cross-linked, pregelatinized potato starch, and esterified, cross-linked, pregelatinized potato starch, wherein the viscosity of the cross-linked, pregelatinized, potato starch is 50 mPa·s or less under measurement conditions where the slurry temperature is 20°C, the

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slurry concentration is 5% by mass, the revolving speed is 60 rpm and the noodles are free of powdered or granular oil or fat or emulsifier.

EFFECT OF LUPIN FLOUR SUBSTITUTION ON THE QUALITY AND SENSORY ACCEPTABILITY OF INSTANT NOODLES BY VIJAY JAYASENA, PEGGY P.Y. LEUNG, SYED M. NASAR-ABBAS. Instant noodles are usually made from wheat flour which is low in protein and dietary fiber contents. To improve the nutritional quality of instant noodles, samples were prepared by incorporating lupin flour, which is high in protein and dietary fiber contents, at 10, 20, 30, 40 and 50% levels. Two sets of noodle samples were made; one with 0.05% added ascorbic acid and the other without added ascorbic acid. The uncooked samples were evaluated for color, protein, fat and mineral contents. The cooked samples were analyzed for cooking loss, color and textural properties in comparison with the control sample. Sensory characteristics (color, flavor, texture, taste and overall acceptability) were also evaluated to find out the optimum acceptable level of lupin flour incorporation in the instant noodles. There was no significant increase in the cooking losses at $\leq 30\%$ lupin incorporation. The color of both uncooked and cooked noodles became more yellowish with the increase in lupin concentration. However, the addition of ascorbic acid (0.05%) improved the color of cooked instant noodles. The addition of 20% lupin flour improved the nutritional value of the product by increasing protein by 42% and dietary fiber by approximately 200% without affecting the sensory properties of instant noodles.

ADDITION OF GREEN BANANA FLOUR TO INSTANT NOODLES: RHEOLOGICAL AND TECHNOLOGICAL PROPERTIES BY MARIA GABRIELA VERNAZAMÁRCIA AROCHA GULARTEYOON KIL CHANG. Nowadays, there is considerable concern about the consumption of high fat and low fiber foods. The objective was to develop instant noodles (IN) with functional properties by adding 10% green banana flour (GBF), and determine the effect of this on the dough rheological properties. Noodles were obtained through sheeting, steaming and frying process. The IN were characterized for their moisture, fat, resistant starch (RS), texture and color. The proximate composition showed that GBF was rich in ash (3.16%), fiber (8.88%) and RS (45.70%), being an attractive raw material for the production of functional IN. When 10% GBF was added, rheological properties of dough changed because of dilution of gluten, making it weaker, but with enough extensibility and elasticity for functional IN production. The addition of 10% GBF significantly reduced the oil content in all the frying times, when compared to the control noodles, increasing also RS content. Ban10 noodles showed a lower L^* value (78.12), indicating a darker color than the control noodles (92.36). In analyzing the firmness of Ban10 noodles, a decrease was observed due to dilution of gluten. However, Ban10 noodles showed a better nutritional profile due to the increase in ash and dietary fiber contents. In conclusion it can be affirmed that Ban10 noodles can be considered as functional IN, due to the presence of fiber (5.79%) and RS (3.98%) contents in the final product.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

The product development studies include product formulation and utilization, shelf-life testing, nutrient-content and microbial analysis and package development.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The study will be using a score card as a research instrument. It looked into the four (4) sensory qualities such as appearance, aroma, taste and texture as the variables. In scoring the variables of the study, the Nine Point Hedonic Scale using a score card was used in evaluating the product.

Statistical Package for Social Sciences (SPSS) is used to generate and processed all the data needed in the study. Mean is used to determine the acceptability of the four sensory qualities tested of the products. The ANOVA was used in order to analyze the differences of the products in four sensory qualities among three treatments.

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Three trials will be done to come up with the most acceptable product in terms of sensory qualities such as appearance, aroma, taste, and texture. The finished product will be evaluated with the panel of experts for sensory evaluation and the acceptability will be evaluated by the consumers.

Product Formulation and Utilization

Sopa de Fideo à la Arroz Caldo

Chicken Broth: Saute the onion, garlic, and ginger until golden brown. Add water, lemongrass, atsute, black pepper powder, salt, chicken. Bring to boil until the chicken meat becomes tender. Get the chicken and separate the meat from the bones. Set aside the chicken meat and return the bones to the broth. Simmer until the broth achieves its desired flavor.

Noodle: Blend the chicken meat, egg, water (broth), onion, and garlic. Combine the flour, glutinous rice, cornstarch, and salt. Make a wheel in the center and add the wet ingredients and form a dough. On the kneading table, sprinkle an ample amount of flour. Place dough onto the floured table. Knead the dough until it is smooth and elastic. Divide it into equal portions and roll the dough using a rolling pin until flattened. Using a noodle maker passes the dough to acquire the desired thickness and consistency of the size of the noodles. Boil the noodles until cooked and shock with cold water. Set Aside.

Chicken Balls: Combine the flour, egg, chicken meat, salt, black pepper powder, and bread crumbs. Form a ball shape using two spoons. Heat the oil until the temperature for frying achieve. Fry the chicken balls until golden brown. Place the fried chicken balls in a paper towel to reduce the oil. Set aside.

Sopa de Fideo à la Arroz Caldo: Place cooked noodles in a bowl and add the hot broth. Arrange the chicken balls, sliced egg, and garnished with chopped onion leaves and toasted garlic. Serve.

Sopa de Fideo à la Chicken Tinola

Chicken Broth: Saute the onion, garlic, and ginger until golden brown. Add water, lemongrass, salt, and chicken. Bring to boil until the chicken meat becomes tender. Get the chicken and separate the meat from the bones. Set aside the chicken meat and return the bones to the broth. Simmer until the broth achieves its desired flavor.

Noodle: Blend the chicken meat, egg, water (broth), onion, and garlic. Combine the flour, glutinous rice, cornstarch, and salt. Make a wheel in the center and add the wet ingredients and form a dough. On the kneading table, sprinkle an ample amount of flour. Place dough onto the floured table. Knead the dough until it is smooth and elastic. Divide it into equal portions and roll the dough using a rolling pin until flattened. Using a noodle maker passes the dough to acquire the desired thickness and consistency of the size of the noodles. Boil the noodles until cooked and shock with cold water. Set Aside.

Green Papaya and Malunggay Leaves: Wash the green papaya and malunggay leaves. Slice the papaya into strips. Sort out the malunggay leaves. Bring to boil in the broth the green papaya and separate once it is cooked. Blanch the malunggay leaves for 1 minute. Set aside.

Sopa de Fideo à la Chicken Tinola: Place cooked noodles in a bowl and add the hot broth. Arrange the chicken sliced, egg, cooked green papaya and garnished with malunggay leaves. Serve.

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(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	2023											
	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
1. Gathering and reading materials connected to the study of interest												
2. Writing of Proposal												
3. Product Formulation and Development												
4. Product Evaluation												
5. Nutrient Content, Microbial Analysis and BFAD approved												
6. Analysis of Findings and writing of Data Gathered												
7. Colloquium /Defenses												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The expected output of this study will be the Development of Sopa de Fedio (Noodle Soup) Ala Arroz Caldo and Ala Chicken Tinola.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The quality standards, cooking guide, and other IEC materials will be developed and produced. It will be disseminated and distributed to various interested stakeholders.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this study will be the following: chicken grower, noodle makers or producers,vegetable growers, homemakers, children, food enthusiast, and sausage vendors.

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(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Dr. Ace D. Barredo	Project leader	Foods

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Sources of Expenses	Quantity	Amount
A. Ingredients (In Three Trials in three treatments) <u>Sopa de Fideo à la Arroz Caldo</u> Chicken Broth : garlic, ginger, onion, lemongrass, atsute, oil, black pepper powder, salt, chicken and water Noodle: flour, glutinous rice, constarch, salt, water (broth), chicken meat, egg, onion, garlic Chicken Balls: flour, egg, chicken meat, salt, black pepper powder and bread crumbs <u>Sopa de Fideo à la Chicken Tinola</u> Chicken Broth : garlic, ginger, onion, lemongrass, oil, salt, chicken and water Noodle: flour, glutinous rice, constarch, salt, water (broth), chicken meat, egg, onion, garlic Green Papaya and Malunggay leaves		Php 20,000.00
B. Materials, Tools, and Equipment		
LPG refill	2 tanks	2,000.00

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Bond paper substance 20 8x11 ½	5 rims	930.00
Printer ink assorted colors	3 sets	2,370.00
Photo paper	10 sets	1,200.00
C. Nutrient-Content, Microbial Analysis, and BFAD Approved, Shelf-life test		40,000.00
D. Packaging		10,000.00
E. English Editor's Fee		5,000.00
F. Publication		20,000.00
G. Presentation		20,000.00
H. Miscellaneous		5,000.00
Total Expenses		Php 126,500.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from **CAPSU Research Journal**, <http://researchjournal.capsu.edu.ph/>)

Noodle | Definition of noodle in English by Oxford Dictionaries". *Oxford Dictionaries | English*. Retrieved 7 May 2019.

Syeda JuveriyaFathimallaiyarajaNallamuthuFarhathKhanum. 12 - Vitamins and minerals fortification using nanotechnology: bioavailability and Recommended Daily Allowances.
<https://www.sciencedirect.com/science/article/pii/B9780128043042000123>

Instant-Noodles-Health-Impact retrieved from parkwayeast.com.sg/healthplus/article

Nagayama Yoshiaki +Sanyo Foods Co Ltd + Nz733059 (A). Dried Noodles And Production Method Thereof.<https://Worldwide.Espacenet.Com>

Effect Of Lupin Flour Substitution On The Quality And Sensory Acceptability Of Instant Noodles By Vijay Jayasena,Peggy P.Y. Leung,Syed M. Nasar-Abbas.
<https://doi.org/10.1111/J.1745-4557.2010.00353.X>

Addition Of Green Banana Flour To Instant Noodles: Rheological And Technological Properties By Maria Gabriela Vernazamárchia Arocha Gularteyoon Kil Chang.
<https://www.scielo.br/J/Cagro/A/Nhprzrkk6yc3fqwpwjy7h/?Lang=En&Format=Html>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

ACE DISTOR BARREDO, was born on June 07, 1988 at Roxas Memorial Provincial Hospital, Roxas City, Capiz. He spent his elementary years at Plaridel East Elementary School, and his secondary education at Capiz National High School. He obtained his bachelor's degree at Capiz State University with the degree of Bachelor of Science in Industrial Education major in Foods as Cum-luade. He finished his master's and doctorate degree at the same university. Presently, he is an Assistant Professor I handling major and professional subjects under

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